

ROHAN BOLLE

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EDUCATION

B.S. Georgia Southern University, Physics and Astronomy *May 2026*
Minor in Mathematics

PUBLICATIONS

R. Bolle, I. Jarra, and J. A. Secrest, “An Exposition on the Kaniadakis κ -Deformed Decay Differential Equation,” *Mathematical and Computational Applications*, 30(5), 115 (2025).
<https://doi.org/10.3390/mca30050115>

Rohan Bolle, Ibrahim Jarra, Jeffery A. Secrest et al. “Scale-Dependent Gravity in Quantum Cosmology I: Singularity Resolution and Semiclassical Dynamics”
(*Yet to be published*)

CONFERENCE PRESENTATIONS

Poster presentation: The Southeastern Section of the American Physical Society, Oxford, MS. “Gravitational Phase Transition in the Early Universe,” November 2022.

Oral Presentation: “Gravitational Phase Transitions in Cosmological Models with a Scale-Factor-Dependent Gravitational Constant,” GS4 Symposium, Georgia Southern University (2026).

HONORS/AWARDS

Bob and Janice Pound Space Exploration & Aerospace Endowed Scholarship *Fall 2023*
Georgia Southern University

Georgia Southern Physics Research Scholar *Summer 2025*

Sigma Pi Sigma, the National Physics Honor Society *Spring 2026*

International Diversity Scholarship *2024 & 2025*
Georgia Southern University

President’s List *Fall 2024, Spring 2024, & Spring 2021*
Georgia Southern University

Dean’s List *Spring 2022*
Georgia Southern University

RESEARCH EXPERIENCE

Undergraduate Research Assistant *August 2022 - Present*
Department of Physics, Georgia Southern University

- Studied a modified mathematical model of decay based on **Kaniadakis κ -deformed statistics**, using analytical derivations and numerical methods to characterize its behavior and compare it with the **standard exponential decay model**.

- Derived the **Friedmann and de Sitter cosmological equations** and extended them to incorporate a scale-factor-dependent gravitational parameter **$G(a)$** ; implemented the resulting models numerically in **Python** to investigate gravitational phase transitions and early-universe dynamics.
- Conducted parameter-space studies and generated and interpreted **numerical solutions** for the **scale factor**, **Hubble parameter**, and **effective energy density** to characterize model behavior and identify physically viable cosmological scenarios.
- Collaborated with research advisor and student peers on interpreting results and preparing material for **conference presentations** and **peer-reviewed publications**.

TEACHING AND MENTORING EXPERIENCE

Math and Physics Tutor

Smart Center, Georgia Southern University

May 2021 to May 2026

- Tutor students in **Calculus I–III and University Physics I–II**, averaging over **150 students** per semester.
- Develop study guides, problem sets, and exam review materials to support conceptual understanding and problem solving.
- Teach mathematical, organizational, and problem-solving strategies in a structured, collaborative environment.

Teaching Assistant

Georgia Southern University

January 2026 to May 2026

- Assisted with instruction for undergraduate **Physics I laboratory** sections of approximately **40 students** twice weekly and assisted with laboratory preparation, organization of instructional materials, and weekly lab setup
- Guided students through experiments, laboratory procedures, data collection, and analysis while reinforcing course concepts.
- **Evaluated and graded laboratory** reports in a timely manner while maintaining accurate academic records.

Science Tutor and Mentor

Remote, Akshaya Foundation

May 2020 - August 2020

- Volunteered to provide daily remote tutoring in **high school Physics and Chemistry** to under-resourced students in India during the **COVID-19** pandemic.

Private Tutor

Savannah, GA

August 2023 - Present

- High School Math and Physics Tutor, AP, and Honors Physics

PROFESSIONAL AFFILIATIONS

Media Team Director

SciTeens

August 2021- 2023

- Managed and monitored social media platforms to promote **STEM accessibility** and outreach.
- Led a **team of five**, delegating tasks and organizing monthly meetings to plan campaigns and content.
- Produced and edited videos using **Adobe Premiere Pro** and graphics using Adobe Photoshop for YouTube, Instagram, and Facebook.

Armstrong Representative

Society of Physics Students, Georgia Southern University

August 2022 - 2024

- Organized workshops for physics majors attended workshops such as LaTeX and Gnu plot.
- Coordinated bi-weekly meetings for physics majors on the Armstrong campus.